Exact Singularities of Homogeneous Ricci Flow on Products of Round Spheres

Route-A source-local certificate HCS-C281
Revision round 0

1 September 2026

Abstract

For arbitrary $m \geq 1$, dimensions $d_i \geq 1$, and scales $a_i > 0$, we solve the unnormalized Ricci flow of the product metric $\bigoplus_i a_i g_{S^{d_i}}$ exactly. Every tied first-collapse factor is retained, yielding sharp scalar-curvature, full-curvature, volume, diameter, and Type-I blowup laws. This is a nonlinear metric-evolution theorem, not a heat-trace, spectral-zeta, or determinant construction.

1 Frozen geometric owner

Let

$$M = \prod_{i=1}^{m} S^{d_i}, \qquad d_i \in \mathbb{Z}_{\geq 1}, \qquad g(0) = \bigoplus_{i=1}^{m} a_i g_i, \quad a_i > 0, \tag{1}$$

where g_i is the unit round metric. Put $c_i = d_i - 1$ and $n = \sum_i d_i$. We use the unnormalized equation

$$\partial_t g = -2\,\mathrm{Ric}(g) \tag{2}$$

with its physical clock. Hamilton's original program fixes (2) as a geometric evolution equation [1]; Chow–Knopf provide standard special-geometry, scaling, and singularity conventions [2]. Those sources are cited only for equation lineage and vocabulary. All product, endpoint, and normalization claims below are derived here.

For $d_i \geq 2$ define $T_i = a_i/(2c_i)$; for $d_i = 1$ define $T_i = \infty$. This convention is essential: a circle is Ricci-flat, not a factor with an undefined clock.

Theorem 1 (exact flow and maximal interval)**.** *The solution of (2) is*

$$g(t) = \bigoplus_{i} a_i(t) g_i, \qquad a_i(t) = a_i - 2c_i t. \tag{3}$$

If some $c_i > 0$, its maximal Riemannian interval is $(-\infty, T)$, where $T = \min_i T_i$. If every $d_i = 1$, (3) is stationary for all $t \in \mathbb{R}$.

Proof. As a $(0,2)$ tensor, the Ricci tensor is unchanged by constant metric rescaling. The unit round factor satisfies $\mathrm{Ric}(g_i) = c_i g_i$, and the Ricci tensor of a Riemannian product is the direct sum of factor tensors. Hence (2) is exactly $a_i' = -2c_i$, proving (3). The coefficients remain positive precisely on the stated interval. The displayed solution solves the full equation; compact Ricci-flow uniqueness identifies it with the ambient solution while it exists. □

2 Closed geometric observables

Mixed sectional curvatures vanish, while the ith factor has sectional curvature $a_i(t)^{-1}$. Consequently

$$R(t) = \sum_i \frac{d_i c_i}{a_i(t)}, \qquad |\,\mathrm{Ric}\,|^2(t) = \sum_i \frac{d_i c_i^2}{a_i(t)^2}, \tag{4}$$

$$|\,\mathrm{Rm}\,|^2(t) = \sum_i \frac{2 d_i c_i}{a_i(t)^2}, \qquad \frac{V(t)}{V(0)} = \prod_i \left(\frac{a_i(t)}{a_i}\right)^{d_i/2}. \tag{5}$$

Here $V(t) = \mathrm{Vol}(M, g(t))$. Differentiation gives

$$\frac{d}{dt} \log V(t) = -R(t). \tag{6}$$

The product distance also gives the exact diameter

$$\mathrm{diam}(M, g(t)) = \pi \sqrt{\sum_i a_i(t)}. \tag{7}$$

The initial product is Einstein with positive constant λ exactly when every factor is curved and $c_i/a_i = \lambda$ for every i. Equivalently, all finite clocks coincide at $T = (2\lambda)^{-1}$, and then

$$g(t) = (1 - t/T)g(0). \tag{8}$$

If all factors are circles, the product is instead Ricci-flat for arbitrary scales. A mixture of flat and curved factors cannot be Einstein.

References

[1] Richard S. Hamilton, *Three-manifolds with positive Ricci curvature*, Journal of Differential Geometry 17 (1982), 255–306, doi:10.4310/jdg/1214436922.

[2] Bennett Chow and Dan Knopf, *The Ricci Flow: An Introduction*, Mathematical Surveys and Monographs 110, American Mathematical Society, 2004, doi:10.1090/surv/110.